

CENTRIFUGAL PUMP P-104 — OEM OPERATION & MAINTENANCE MANUAL

Model: CP-250 Series | Serial: SN-20210847 | Manufacturer: FlowTech Industries

Revision: Rev 3 | Date: January 2025 | Equipment ID: P-104

1. Equipment Overview

Pump P-104 is a horizontal centrifugal pump designed for continuous process duty in Unit-3 of the refinery. Nominal capacity: 85 m³/h at 4.2 bar discharge pressure. Design speed: 1480 RPM. Motor drive: M-104 (75 kW, 415V, 3-phase).

2. Normal Operating Parameters

The following parameters define healthy operation for Pump P-104:

Parameter	Normal Range	Warning Threshold	Critical Threshold
Vibration (mm/s)	< 2.3	2.3 – 4.5	> 7.1 (ISO Zone C)
Bearing Temp (°C)	55 – 70	70 – 85	> 85
Discharge Pressure	3.8 – 4.5 bar	< 3.0 bar	< 2.0 bar
Flow Rate (m³/h)	80 – 95	65 – 80	< 60
Motor Current (A)	120 – 135	135 – 145	> 145

3. Maintenance Schedule

3.1 Daily Checks: Verify vibration readings, bearing temperatures, seal leak-off flow, and discharge pressure against operating limits defined in Section 2.

3.2 Weekly: Inspect mechanical seal for leakage. Check coupling alignment. Verify motor insulation resistance > 100 MΩ.

3.3 Lubrication: Re-lubricate bearings every 500 operating hours using Mobil Grease XHP 222 (Shell Alvania EP2 equivalent). Over-lubrication causes overheating — use exactly 15g per bearing housing. Lubrication interval must not exceed 500 hours under any circumstances.

3.4 Annual Overhaul: Full disassembly, bearing replacement, wear ring inspection, impeller clearance check (max 0.4 mm), seal replacement if leakage > 5 ml/hr.

3.2.1 Bearing Maintenance — Critical Procedure

Bearing degradation is the most common failure mode for P-104. Early indicators include vibration increase above 4.5 mm/s and bearing temperature above 75°C. When vibration reaches 7.1 mm/s (ISO 10816 Zone C), immediate shutdown is required.

Historical data: Three bearing failures recorded on P-104 (2022, 2023, 2025). All three were preceded by lubrication interval exceedance of > 40 hours beyond the 500-hour limit. Root cause in all cases: grease breakdown leading to metal-to-metal contact.

Replacement bearing specification: SKF 6205-2RS1/C3 (deep groove ball bearing). Do not substitute with open bearings in this application — seal contamination risk.

4. Failure Modes and Corrective Actions

4.1 High Vibration: Check bearing condition, coupling alignment, impeller balance. If vibration > 7.1 mm/s, isolate immediately. Do not operate above ISO Zone C limit.

4.2 Seal Failure: Increased leak-off flow (> 5 ml/hr) indicates seal wear. Replace mechanical seal — John Crane Type 1 or equivalent. Dry running causes immediate seal failure — never start pump without priming.

4.3 Cavitation: Erratic vibration with crackling noise, reduced flow and pressure. Check suction strainer, reduce speed, or increase suction pressure. Cavitation causes rapid impeller erosion.

4.4 Bearing Failure: Typically preceded by rising vibration over 2–4 weeks. Replace both radial and thrust bearings simultaneously. Never replace only one bearing.

5. Safety Requirements

All maintenance on P-104 must follow OISD-113 (Maintenance of Electrical Installations) and the plant Permit-to-Work system. LOTO (Lock-Out Tag-Out) is mandatory before any mechanical work. Confined space entry requires gas testing per Factory Act Section 7A.